# Can EEG's and dart games be used for training embedded systems and AI?

Yes, **EEG and target games** can train embedded systems and AI models.

**Can EEGs and games be used for training embedded systems and AI?**

Requires at least 5 papers that directly answer your question. Try adjusting your query to find more papers.

FIGURE 1 Consensus on using EEG and target games for training embedded AI systems

Research shows that electroencephalography (EEG) combined with motor-target or serious gaming paradigms effectively trains embedded systems and artificial intelligence across closed-loop feedback, edge computing deployment, and adaptive reinforcement learning (Shin et al., 2022; Nguyen et al., 2025; Kar et al., 2022).

## Closed-Loop Brain-Machine Reinforcement

Brain-computer interfaces (BCIs) paired with target-based and action games supply real-time neural signals that train or adapt algorithmic agents (Kerous et al., 2018; Li et al., 2021). Error-related potentials (ErrPs) and motor intention signals generated during physical or simulated motor tasks serve as reward or correction signals in reinforcement learning environments (Shin et al., 2022; Kar et al., 2022).

- Human error-related potentials from target throws allow a **reinforcement learning automaton** to adapt state-action probabilities for robotic arm arm-control tasks with over 90% success (Kar et al., 2022).
- Closed-loop wearable systems analyze ErrP patterns via deep learning to execute **emergency braking interrupts** or optimize machine decision-making on autonomous platforms (Shin et al., 2022).
- In vitro neural cultures embedded via multielectrode arrays demonstrate **goal-directed learning** within five minutes when subjected to closed-loop game feedback in Pong-like environments (Kagan et al., 2021).
- Single-trial classification using spatial pattern algorithms predicts **human competitive choices** to guide proactive AI systems in real-time interactions (He et al., 2023).

## Embedded Edge AI Deployments

Deploying machine learning models directly onto resource-constrained embedded platforms requires tailored architectures capable of handling noisy, high-dimensional EEG inputs (Nguyen et al., 2025; Mutlu, 2025).

| Platform Architecture | Model & Methods | Target Application | Reported Performance |
|---|---|---|---|
| STM32 MCU Board (Nguyen et al., 2025) | 1D CNN with TinyML (Nguyen et al., 2025) | Real-time robotic platform control (Nguyen et al., 2025) | **99.3% accuracy** across blink classes (Nguyen et al., 2025) |
| Edge AI Accelerator (He et al., 2025) | Wavelet packet decomposition + CNN (He et al., 2025) | Portable closed-loop serious gaming (He et al., 2025) | **21.85% reduction** in theta/beta ratio (He et al., 2025) |
| OpenBCI Cyton Interface (Aume et al., | CNN with transfer learning (Aume et al., | Mechatronic arm and game control (Aume et | Superior **activation accuracy** over KNN/LDA |

| Platform Architecture | Model & Methods | Target Application | Reported Performance |
|---|---|---|---|
| 2023) | 2023) | al., 2023) | (Aume et al., 2023) |
| Wireless In-Ear Module (Shin et al., 2022) | FIR filtering (1–8 Hz) + pretrained DL (Shin et al., 2022) | Brain-AI Closed Loop (BACLoS) (Shin et al., 2022) | Real-time **ErrP verification** for AI actions (Shin et al., 2022) |
| Minimal 5-Channel Headset (Wang et al., 2026) | Transformer with leave-one-subject-out (Wang et al., 2026) | Pediatric adaptive intervention games (Wang et al., 2026) | **95.88% accuracy** in engagement states (Wang et al., 2026) |

FIGURE 2 Comparison of embedded and edge architectures processing neuroelectric signals

## Neuroadaptive State Estimation in Gaming

Serious games evoke cognitive and affective states, generating rich electrophysiological streams that train predictive AI models (Gonzalez-Vazquez et al., 2024; Anwar et al., 2017; Kumar & Kumar, 2021). In obstacle-dodging racing simulations, recurrent neural networks with gated recurrent units classify multi-level mental stress with up to 94% accuracy (Gonzalez-Vazquez et al., 2024). Similarly, deep neural network frameworks categorize distinct emotional valences during five-minute gaming intervals with minimal latency (Kumar & Kumar, 2021).

Biocybernetic control loops also rely on dynamic EEG spectral shifts, such as theta and alpha band modulations, to tune game difficulty and keep players engaged (Ewing et al., 2016). Machine learning classifiers like Naive Bayes can isolate novice versus expert skill levels from as few as four relevant channels in the frontal and occipital lobes (Anwar et al., 2017). Despite these successes, training embedded models remains constrained by low signal-to-noise ratios, inter-subject variance, and the processing limits of low-power microcontrollers (Nguyen et al., 2025; Hamzah & Abdalla, 2024; Mutlu, 2025).

In summary, EEG signals recorded during target, motor, and serious games provide robust training and feedback data for AI models and embedded edge platforms (Gonzalez-Vazquez et al., 2024; Nguyen et al., 2025; Kar et al., 2022).

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